

LabelLex

AI-Powered Legal Metrology Statutory Compliance & Forensic Label Verification Engine

Comprehensive Engineering Documentation & System Specification for Real-Time LMPC Rules 2011 Automated Audit, Multi-Panel OCR Pipeline, and Multi-Stakeholder Enforcement Architecture.

PROBLEM STATEMENT

SIH26034 / Legal Metrology Pre-Screening

PRIMARY WORKING PATH

C:\Projects\Labellex

CORE OCR TECHNOLOGY

Tesseract.js WebAssembly + Canvas GPU
Preprocessor

STATUTORY FRAMEWORK

LM (Packaged Commodities) Rules, 2011 & LM Act, 2009

TARGET STAKEHOLDERS

Consumers, Enforcement, Manufacturers, Officers

STATUS & BUILD VERIFICATION

Production Ready (100% Automated Test Suite
Passing)

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1. Project Overview & Statutory Motivation

1.1 Problem Statement & Background

Under the **Legal Metrology Act, 2009** and the **Legal Metrology (Packaged Commodities) Rules, 2011 [LM(PC) Rules, 2011]**, all pre-packaged commodities distributed across India must prominently display mandatory statutory declarations on their Principal Display Panel (PDP) and Information Panels. Non-compliance results in severe penal liabilities under **Section 36(1)** (fines ranging from ₹25,000 to ₹50,000 and compounding under Section 48).

Traditional verification workflows face critical bottlenecks:

- **Manual Inspection Bottlenecks:** Enforcement officers manually verify tiny font sizes, complex manufacturer addresses, and unit sale price math across thousands of retail SKUs.
- **Consumer Information Asymmetry:** Consumers struggle to verify deceptive packaging, over-stickered Maximum Retail Prices (MRP), or missing mandatory tax disclosures.
- **Pre-Market Artwork Friction:** Manufacturers and packaging design agencies lack instant pre-market compliance tools to validate packaging artwork before incurring expensive mass printing runs.

LabelLex resolves this by providing an autonomous, multi-panel Optical Character Recognition (OCR) and statutory rule verification platform that audits packaging declarations in under 2 seconds.

1.2 Target Stakeholder Personas & Roles

LabelLex integrates 4 dedicated, tailored portals engineered for distinct operational requirements:

PORTAL / ROLE	TARGET AUDIENCE	KEY CAPABILITIES & WORKFLOW
1. Consumer Scanner	Everyday citizens, shoppers, consumer activists	Quick mobile camera scan, instant PASS/FAIL safety verdict, 1-tap grievance filing to National Consumer Helpline with reference ID tracking.
2. Enforcement Crawler	Legal Metrology state departments, E-commerce compliance wings	E-commerce URL scraping pre-screening, high-volume batch artwork scanning, penalty liability calculation (₹25k–₹50k), CSV/JSON evidence export.
3. Manufacturer Tool	FMCG brand owners, packers, packaging design agencies	Pre-market artwork pre-screening, automated Unit Sale Price (USP) calculator, downloadable official Pre-Market Compliance Certificate.
4. Field Officer App	Legal Metrology field inspectors (District/State level)	Mobile inspection tool, multi-panel evidence photo attachment (shelf context, over-stickered price tags), compounding notice drafting, secure PIN login.

1.3 High-Level System Workflow

The user captures or uploads packaging photos (Front, Back, Side, or single panel). The system executes GPU-safe image optimization, runs Tesseract.js client-side OCR across candidate orientations, repairs noisy dot-matrix character anomalies, evaluates the extracted text against 8 statutory LM(PC) rules, computes font prominence and panel placement scores, and renders a forensic audit report with actionable manufacturer communication channels.

2. System Architecture & Request/Response Flow

2.1 Complete Architectural Topology

LabelLex is built upon a high-performance **Client-Side Edge Execution Architecture**. All compute-intensive image processing, neural-based OCR inference, regex matching, and report generation execute directly on the user's client hardware (WebAssembly/Worker threads), ensuring zero cloud operating costs, instant sub-second response times, and total confidentiality of proprietary artwork.

- **Data Privacy & Trade Secrets:** Pre-market packaging artwork never leaves the manufacturer's local device before launch.
- **Offline Field Readiness:** Officers in remote warehouses or retail basements without cellular connectivity can perform instant statutory audits using cached Tesseract language models.

2.3 End-to-End User Transaction Sequence

1. **Slot Ingestion:** User captures packaging panels via `CameraModal` or file picker.
2. **Downscaling & Canvas Normalization:** `imageOptimizer.ts` rescales images to max 2048px width/height and applies contrast normalization.
3. **OCR Worker Execution:** `ocrPipeline.ts` invokes the Tesseract WebAssembly worker, auto-correcting orientation if needed.
4. **Field Evaluation:** `fieldMatcher.ts` maps extracted tokens to the 8 mandatory rules in `rules.ts`.
5. **Verdict & Metric Computation:** Computes USP, Rule 7 prominence ratios, panel placement, and overall score (0–100%).
6. **State Storage & Rendering:** Commits the record to `sharedStore.ts` and renders `ComplianceReport.tsx`.

3. Frontend Subsystem & Component Hierarchy

3.1 Languages, Frameworks & Tooling

- **Core Framework:** React 19 (v19.2.8) with functional components, custom hooks, and StrictMode support.
- **Language:** TypeScript (v6.0.2) with strict type safety, zero `any` leaks in engine contracts, and full interfaces.
- **Build Tool:** Vite 8 (v8.2.2) leveraging ES modules for instant hot module replacement (HMR) and optimized rollup production bundles.
- **Styling Engine:** Tailwind CSS (v3.4.19) configured with custom pastel palettes (mint, amber, coral, purple, charcoal) and smooth micro-interactions.
- **Icons:** Lucide React (v1.37.0) providing contextual statutory icons.

3.2 Directory & Component Mapping

FILE PATH	COMPONENT / MODULE	DESCRIPTION & RESPONSIBILITIES
src/App.tsx	App (Root Router)	Main application orchestrator; handles portal navigation, modal rendering, active report state, and sample package injection.
src/components/Navbar.tsx	Navbar	Top navigation bar with portal switcher, language toggle (EN/HI), user authentication badges, and quick-scan triggers.
src/components/LandingPage.tsx	LandingPage	Interactive dashboard selection grid showcasing the 4 stakeholder portals with feature highlights and stat counters.
src/components/ImageUploadPipeline.tsx	ImageUploadPipeline	Multi-panel image uploader; manages slot state (Front, Back, Side, Single), drag-and-drop, camera invocation, and progress bars.
src/components/CameraModal.tsx	CameraModal	High-resolution camera viewfinder utilizing navigator.mediaDevices.getUserMedia with live frame guides and shutter controls.
src/components/ComplianceReport.tsx	ComplianceReport	Forensic audit report view; displays verdict badge, 8-rule statutory table, Rule 7 prominence cards, Rule 8 placement cards, evidence gallery, and manufacturer dispatch.
src/components/CertificateModal.tsx	CertificateModal	Official Legal Metrology Pre-Market Compliance Certificate view with security seals, QR verification badges, and print styles.
src/components/AuthModal.tsx	AuthModal	Role-based authentication modal supporting Officer badge verification (e.g. LM-MH-8821) and PIN security.
src/components/SampleSelectorModal.tsx	SampleSelectorModal	Pre-loaded test packaging suite (Britannia English Tea Cake, D-Mart Soluble Coffee, California Almonds, etc.) for instant verification.
src/dashboards/ConsumerDashboard.tsx	ConsumerDashboard	Citizen portal with simplified 1-tap scanner and grievance filing triggers.
src/dashboards/EnforcementDashboard.tsx	EnforcementDashboard	E-commerce crawler and batch inspection portal with automated penalty liability scoring.
src/dashboards/ManufacturerDashboard.tsx	ManufacturerDashboard	Artwork pre-compliance certification tool with automated USP generation.
src/dashboards/OfficerDashboard.tsx	OfficerDashboard	Field officer inspection tool with multi-evidence attachment and compounding notice generator.

3.3 State Management & Language Localization

The central reactive store is implemented in `src/store/sharedStore.ts`. It maintains:

- **Current Active Role:** `'consumer' | 'enforcement' | 'manufacturer' | 'officer' | 'landing'`
- **Audit Reports Database:** Array of `ComplianceReportData` records persisted to `localStorage`.
- **Language Context:** `'en' | 'hi'` backed by a dictionary in `src/store/translations.ts` covering 100% of UI strings, legal terms, and guidance notes.

4. Backend Logic Engines & Verification Algorithms

4.1 Statutory Rules Specification Engine (`src/engine/rules.ts`)

Defines the 8 mandatory declarations stipulated under Rule 6(1) of the LM(PC) Rules, 2011:

RULE	CLAUSE NAME	LEGAL REFERENCE	STATUTORY MANDATE & THRESHOLD
Rule 6(1)(a)	Manufacturer / Packer / Importer	Rule 6(1)(a), LM(PC) Rules 2011	Complete physical name and address with PIN code; must include prefixes like "Mfg by:", "Packed by:", or "Marketed by:".
Rule 6(1)(b)	Common / Generic Name	Rule 6(1)(b), LM(PC) Rules 2011	Clear generic identity of the commodity (e.g. "Soluble Coffee Powder", "Cake").
Rule 6(1)(c)	Net Quantity	Rule 6(1)(c) & Rule 12	Expressed in standard SI metric units (g, kg, ml, l, N, U). Non-metric units (lbs, oz) are flagged as non-standard.
Rule 6(1)(d)	Month & Year of Mfg/Packing	Rule 6(1)(d), LM(PC) Rules 2011	Mandatory month and year (e.g. "08/2026", "AUG 2026", "DEC/2025"). Year-only declarations are flagged for review.
Rule 6(1)(e)	Maximum Retail Price (MRP)	Rule 6(1)(e), LM(PC) Rules 2011	Indian Rupees (₹/Rs.) with mandatory statutory disclosure: "Inclusive of all taxes".
Rule 6(1)(f)	Consumer Care Details	Rule 6(1)(f), LM(PC) Rules 2011	Name, address, phone number, and email of grievance officer.
Rule 6(1)(s)	Unit Sale Price (USP)	2021 Amendment	Mandatory for packs > 100g/ml; calculated as MRP divided by net metric weight (rounded to 2 decimal places).
Rule 6(10)	Country of Origin	Rule 6(10), LM(PC) Rules 2011	Explicit declaration: "Country of Origin: India" (or foreign nation if imported).

4.2 Pattern Extraction & Field Matcher (`src/engine/fieldMatcher.ts`)

The field matcher executes advanced regex tokenization, character-level OCR repair, and domain-specific heuristics. Key algorithms include token sanitization, structured number correction (repairing 'O' to '0' and 'S' to '5' in numeric contexts), and statutory clause extraction.

4.3 4-Tier Compliance Status Classification Algorithm

Each field is evaluated into one of 4 official categories conforming to SIH26034 requirements:

- VALID** : Fully conforms to statutory rule, metric standard, and legal phrasing.
- NON-STANDARD** : Value is present but uses non-metric units (e.g. "5 lbs") or misses the mandatory tax disclosure (e.g. "MRP ₹120" without "incl. of all taxes").
- REVIEW** : Text is ambiguous, partially obscured, or requires manual human officer verification.
- VIOLATION** : Mandatory statutory clause is completely missing from all scanned panels.

5. Database, Persistence & Data Schemas

5.1 Comprehensive TypeScript Data Models

Data structures are strictly defined in `src/engine/rules.ts`:

- `ComplianceReportData`: Master audit entity containing product metadata, overall score, 8 field evaluations, readability assessment, placement assessment, and supporting evidence photos.
- `FieldEvaluationResult`: Clause-level result containing ruleId, extracted text, normalized value, status, confidence, and legal guidance.
- `SupportingEvidence`: In-field evidentiary photos (shelf placement, over-stickered price tags) with timestamp and officer ID.

5.2 Storage Engine & Privacy Rationale

Reports are serialized into the browser's `localStorage` database under the key `labellex_audit_reports_v2`. This ensures immediate persistence across page reloads without requiring an external database cluster. Users can export reports into structured JSON or CSV spreadsheets at any time.

6. Label Scanning & Multi-Panel OCR Pipeline (Deep Dive)

6.1 7-Stage End-to-End Scan Pipeline

The scanning engine in `ocrPipeline.ts` and `imageOptimizer.ts` processes raw camera frames through a 7-stage pipeline:

1. **Image Ingestion:** Ingests File/Blob from camera stream or gallery picker. Validates against 10MB size ceiling.
2. **GPU Canvas Rescaling:** Clamps image dimensions to 2048px maximum width/height to prevent mobile browser WebGL/Canvas memory crashes.
3. **Auto-Orientation Analysis:** Tests 0°, 90°, 180°, and 270° candidate angles against packaging vocabulary keyword density (`scorePackagingKeywords`) to automatically orient sideways or upside-down jars.
4. **WASM OCR Execution:** Spawns a singleton Tesseract.js worker with `PSM.AUTO` page segmentation mode.
5. **Bounding Box & Font Extraction:** Extracts individual line bounding boxes (`x0, y0, x1, y1`) and line pixel heights.
6. **Character & Dictionary Normalization:** Executes `correctStructuredNumbers` to eliminate OCR confusion between numbers and letters.
7. **Multi-Panel Aggregation:** Merges text streams across Front, Back, and Side panels, tagging declarations with their source panel location.

6.2 Multi-Angle Quadrant Auto-Orientation Algorithm

The pipeline tests candidate rotation angles against a dictionary of 40+ statutory keywords (e.g. *manufactured, marketed, net quantity, mrp, inclusive, batch*) to select the optimal angle before extracting tokens.

6.3 Rule 7 Font Size & Rule 2(h)/8 Placement Assessments

- **Rule 7 Readability Assessment:** Computes the pixel height of mandatory declarations relative to the tallest reference brand header. Ratios > 25% are marked `prominent`, 15%–25% `acceptable`, and < 15% `concern`.
- **Rule 2(h) & Rule 8 Placement Assessment:** Verifies that Net Quantity, Commodity Name, and MRP are on the Principal Display Panel (PDP / Front), while manufacturer addresses and customer care are on the Information Panel (Back/Side).

7. Third-Party Tools, Libraries & Dependencies

LIBRARY / TOOL	VERSION	FUNCTION / ROLE	TECHNICAL RATIONALE & INTEGRATION DETAILS
tesseract.js	7.0.0	Client-Side OCR Inference	Compiled WebAssembly port of Google's Tesseract C++ engine. Enables offline text recognition in browser threads.
react / react-dom	19.2.8	UI Component Framework	Declarative component model with high-performance virtual DOM diffing and StrictMode concurrency.
vite	8.2.2	Next-Gen Frontend Tooling	Rollup-based bundler providing sub-second HMR and tree-shaken static asset compilation.
tailwindcss	3.4.19	Utility-First CSS Engine	Zero-runtime styling framework compiled with PostCSS and Autoprefixer.
lucide-react	1.37.0	Statutory Icon Suite	Lightweight, accessible SVG icon library providing contextual icons for all 8 statutory rules.
canvas-confetti	1.9.4	User Experience Feedback	Canvas-based particle animation triggered when packaging achieves 100% PASS compliance score.
typescript	6.0.2	Static Typing & Tooling	Guarantees compile-time type safety across all legal rules, OCR interfaces, and store actions.

8. Data Flow Architecture Diagram

[Physical Package Label]



1. Ingestion: Camera Capture / Multi-Panel File Upload



2. Preprocessing: Canvas GPU Clamp (2048px) + Contrast



3. Multi-Angle OCR: Tesseract WASM (0°, 90°, 180°, 270°)



4. Text Normalization: Structured Numeric Repair (0->0)



5. Rule Verification: LMPC Rules 2011 Regex & USP Math



6. Readability & Placement: Rule 7 Prominence + Rule 8



7. Output: Forensic Audit Report + PDF + Email/WhatsApp

9. Deployment & Infrastructure

- **Production Build Command:** `npm run build` (executes `tsc -b && vite build`).
- **Build Output:** Minified, production-ready static assets in `dist/` (1.20 kB HTML, 42.4 kB CSS, 471 kB JS).
- **Zero-Backend Static Hosting:** Can be deployed directly to Cloudflare Pages, Vercel, Netlify, AWS S3, or GitHub Pages.
- **Live Public Tunneling:** Integrated Cloudflare Tunnel daemon (`cloudflared.exe`) enabling instant zero-config public tunneling for mobile device testing.

10. Key Engineering Challenges & Technical Solutions

CHALLENGE FACED	TECHNICAL ROOT CAUSE	ENGINEERED SOLUTION IN LABELLEX
1. Dot-Matrix & Inkjet Print Degradation	Batch dates and MRP printed using low-res dot-matrix printers on curved jars caused OCR character fragmentation (e.g. reading '0' as 'O' or '5' as 'S').	Engineered <code>correctStructuredNumbers()</code> in <code>fieldMatcher.ts</code> which dynamically repairs numbers flanked by currency symbols or date delimiters.
2. Mobile Browser GPU Memory Crashes	High-resolution 48MP/108MP mobile camera photos exceeded Safari/Chrome iOS canvas memory limits.	Implemented hardware-safe canvas scaling clamping max dimensions to 2048px in <code>imageOptimizer.ts</code> .
3. Sideways & Inverted Jar Scans	Users frequently scan cylindrical cans and jars sideways, causing OCR engines to output complete gibberish.	Implemented 4-quadrant rotation scoring in <code>ocrPipeline.ts</code> testing candidate angles against statutory packaging keywords.
4. Mandatory Tax Disclosure Detection	Manufacturers frequently omit "Inclusive of all taxes", creating Section 36(1) violations.	Built strict regex parser in <code>fieldMatcher.ts</code> that marks MRP without statutory tax disclosure as <code>NON-STANDARD</code> .

11. Comprehensive Tech Stack Summary Table

LAYER / DOMAIN	TECHNOLOGIES & LIBRARIES	ROLE & KEY DELIVERABLE
Frontend Core	React 19.2.8, TypeScript 6.0.2, Vite 8.2.2	Reactive single-page web application with strict type contracts.
Styling & UI	Tailwind CSS 3.4.19, Lucide React 1.37.0	Utility-first responsive design with custom pastel tokens and statutory icons.
Computer Vision & OCR	Tesseract.js 7.0.0, HTML5 Canvas 2D API	Client-side WebAssembly OCR engine with multi-angle rotation heuristics.
State & Persistence	Zustand-style reactive store, Web Storage API	Client-side reactive state management with localStorage audit persistence.
Localization	Custom Dictionary (translations.ts)	100% bilingual support for English and Hindi.
Testing & Quality	TSX (TypeScript Execute), Node.js Test Suites	Automated unit and integration test suites validating real product ground-truth.
Report Dispatch	HTML5 Blob, Mailto URI, WhatsApp API	Instant PDF report generation and 1-tap manufacturer communication.

12. Future Scope & Roadmap

- Native Mobile Applications:** Packaging as iOS and Android native apps using Capacitor / React Native with ONNX Runtime for on-device neural text detection.
- National Consumer Helpline (NCH) API Integration:** Direct automated dispatch of consumer grievances to the government's INGRAM portal.
- 22 Official Indian Languages:** Expanding OCR language models to support regional scripts (Devanagari, Tamil, Telugu, Kannada, Bengali, Gujarati).
- GS1 Barcode & QR Code Cross-Verification:** Instant cross-check between OCR label text and GS1 India DataKart databases to detect counterfeit barcodes.